

# Math Class 9th Federal Board Chapter 01

## Real Numbers Revision and Full Test

| Content Domain / Area         | Topics       | SLO No./ Description                                                                                                                                                                                                      | Form of Assessment | Cognitive Level (Knowledge, Understanding, Application) | Remarks                                                                                                         |
|-------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Domain A: Numbers and Algebra | Real Numbers | [SLO: M-09-A-01]: Explain, with examples, that civilizations throughout history have systematically studied living things [e.g., the history of numbers from Sumerians and its development to the present Arabic system.] | Formative          | K                                                       | Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice. |
|                               |              | [SLO: M-09-A-02]: Describe the set of real numbers as a combination of rational and irrational numbers.                                                                                                                   | Formative          | K                                                       | Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice. |
|                               |              | [SLO: M-09-A-03]: Demonstrate and verify the properties of equality and inequality of real numbers.                                                                                                                       | Formative          | K                                                       | Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice. |
|                               |              | [SLO: M-09-A-04]: Apply laws of indices to simplify radical expressions.                                                                                                                                                  | Summative          | A                                                       | Question(s) will be asked in the annual examination.                                                            |
|                               |              | [SLO: M-09-A-05]: Express a number in scientific notations and vice versa.                                                                                                                                                | Summative          | U                                                       | Question(s) will be asked in the annual examination                                                             |

| Question | سوال                                                                           | A                  | B                     | C                     | D                  |
|----------|--------------------------------------------------------------------------------|--------------------|-----------------------|-----------------------|--------------------|
| 1.       | $\left(\frac{1}{9}\right)^{-\frac{3}{2}} =$                                    | 27                 | $\frac{1}{27}$        | $\pm 27$              | $\pm \frac{1}{27}$ |
| 2.       | The scientific notation of 0.00567 is:<br>0.00567 کی سائنٹفک نوٹیشن کیا ہے گی؟ | $5.67 \times 10^3$ | $5.67 \times 10^{-3}$ | $5.67 \times 10^{-2}$ | $5.67 \times 10^2$ |

$$1. \left(\frac{1}{9}\right)^{-\frac{3}{2}} = \left(\frac{9}{1}\right)^{\frac{3}{2}} = (3^2)^{\frac{3}{2}}$$

$$\text{Rule: } \left(\frac{a}{b}\right)^m \Rightarrow \left(\frac{b}{a}\right)^{-m}$$

$$= 3^{2 \times \frac{3}{2}} = 3^3 = 3 \times 3 \times 3 = 27$$

2-  $0.00567$

$\downarrow$

$5.67 \times 10^{-3}$

$\uparrow \dots \times 10^n$

$$(5.67 \times 10)$$

|                                                                                                        |          |       |                   |                   |
|--------------------------------------------------------------------------------------------------------|----------|-------|-------------------|-------------------|
| 1. Which of the following is equal to $(x^{\frac{2}{3}})^9$ ?<br>کس کے برابر ہے؟ $(x^{\frac{2}{3}})^9$ | $x^8$    | $x^6$ | $x^{\frac{4}{9}}$ | $x^{\frac{3}{2}}$ |
| 2. $5.67 \times 10^4$ in standard form is:<br>$5.67 \times 10^4$ کو معیاری صورت میں ظاہر کریں:         | 0.000567 | 0.567 | 567000            | 56700             |

$(\frac{2}{3})^9$   
 $x^{\frac{2}{3} \times 9} = x^6$   
 $5.67 \times 10^4$   
 $5.67000 \times 10^4$   
 $56700.0 \dots$   
 $5.6 \times 10$   
 $56$

$1.2$   
 $1.20$   
 $1.200$   
 $1.2000$

|                                                                             |                     |                     |                        |                        |
|-----------------------------------------------------------------------------|---------------------|---------------------|------------------------|------------------------|
| 1. $(\frac{1}{4})^{-\frac{1}{2}} =$                                         | $\frac{1}{2}$       | 2                   | $\sqrt{2}$             | 16                     |
| 2. The scientific notation of 345.6 is:<br>345.6 کی سائنٹیفک نوٹیشن کیا ہے؟ | $3.456 \times 10^2$ | $3.456 \times 10^3$ | $3.456 \times 10^{-2}$ | $3.456 \times 10^{-3}$ |

$$345.6$$

|                                                                                             |                   |                   |                   |                   |
|---------------------------------------------------------------------------------------------|-------------------|-------------------|-------------------|-------------------|
| The expression $\sqrt[3]{a^5}$ equals:<br>ریڈیکل $\sqrt[3]{a^5}$ کی قوت نمائی شکل کیا ہوگی؟ | $a^{\frac{3}{5}}$ | $a^{\frac{5}{2}}$ | $a^{\frac{5}{3}}$ | $a^{\frac{3}{2}}$ |
|---------------------------------------------------------------------------------------------|-------------------|-------------------|-------------------|-------------------|

The expression  $\sqrt[3]{a^5}$  equals.

ریڈیکل  $\sqrt[3]{a^5}$  کی قوت نمائی شکل کیا ہوگی؟

$a^{\frac{3}{5}}$

$a^{\frac{5}{2}}$

$a^{\frac{5}{3}}$

$a^{\frac{3}{2}}$

③  $\rightarrow ( )^{\frac{1}{3}}$   $\rightarrow$  cube root

$$\sqrt[3]{a^5} = (a^5)^{\frac{1}{3}} = a^{5 \times \frac{1}{3}} = a^{\frac{5}{3}}$$

(5 x 1/3)

|     |                                                                             |                                       |                          |                       |                                        |
|-----|-----------------------------------------------------------------------------|---------------------------------------|--------------------------|-----------------------|----------------------------------------|
| i.  | The radical form of $x^{-\frac{3}{2}}$ is:                                  | <del><math>\sqrt[3]{x^2}</math></del> | $\frac{1}{\sqrt{x^3}}$ ✓ | $\sqrt{x^3}$          | $\frac{1}{\sqrt[3]{x^2}}$ <del>X</del> |
| ii. | Which of the given options represents the scientific notation of $0.25^2$ ? | $625 \times 10^{-4}$                  | $62.5 \times 10^{-3}$    | $6.25 \times 10^{-2}$ | $0.625 \times 10^{-1}$                 |

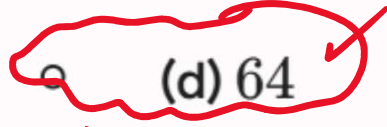
$$x^{-3 \times \frac{1}{2}} = (x^{-3})^{\frac{1}{2}}$$

$$\frac{1}{x^3} = \sqrt{x^{-3}} = \sqrt{\frac{1}{x^3}} = \frac{1}{\sqrt{x^3}}$$

$$\frac{x^{-3} \times x^3}{x^3} = \frac{x^{-3+3}}{x^3} = \frac{x^0}{x^3} = \frac{1}{x^3}$$

1.  $\left(\frac{1}{16}\right)^{-\frac{3}{2}} =$

- (a)  $\frac{1}{4}$
- (b) 4
- (c) 16
- (d) 64



2. 4500 in scientific notation:

- (a)  $4.5 \times 10^3$
- (b)  $45 \times 10^3$
- (c)  $4.5 \times 10^2$
- (d)  $0.45 \times 10^5$

4500. (Handwritten red arrows pointing to the digits 4, 5, and the decimal point)

(i)  $(27x^{-1})^{-\frac{2}{3}}$  .....

✓ (a)  $\frac{\sqrt[3]{x^2}}{9}$

✗ (b)  $\frac{\sqrt{x^3}}{9}$

✗ (c)  $\frac{\sqrt[3]{x^2}}{8}$

✗ (d)  $\frac{\sqrt{x^3}}{8}$

$(27x^{-1})^{-\frac{2}{3}}$

$\left(\frac{27}{x}\right)^{-\frac{2}{3}} = \left(\frac{x}{27}\right)^{\frac{2}{3}}$

~~27~~ 27

2

2/3



$$= \sqrt{\frac{(6^3)^{\frac{2}{3}} \times (5^2)^{\frac{1}{2}}}{\left(\frac{0.04}{100}\right)^{-\frac{1}{2}}}}$$

|   |     |
|---|-----|
| 2 | 108 |
| 2 | 54  |
| 3 | 27  |
| 3 | 9   |
| 3 | 3   |
| 3 | 1   |

$$= \left( \frac{6^{\cancel{2} \times \frac{2}{\cancel{3}}} \times 5^{\cancel{2} \times \frac{1}{\cancel{2}}}}{\left(\frac{1}{25}\right)^{-\frac{1}{2}}}\right)^{\frac{1}{2}}$$

$$= \left( \frac{6^2 \times 5}{(25)^{1/2}} \right)^{\frac{1}{2}}$$

$$= \left( \frac{36 \times 5}{\cancel{5^{\cancel{2} \times \frac{1}{\cancel{2}}}}} \right)^{\frac{1}{2}}$$

$$= \left( 36 \times 5 \right)^{1/2}$$

$$= \left( \frac{36 \times 4}{5} \right)$$

$$= (6^2)^{\frac{1}{2}} = 6^{2 \times \frac{1}{2}}$$

Simplify

$$= \textcircled{6} A$$

$$\frac{(81)^n \cdot 3^5 - (3)^{4n-1} (243)}{(9^{2n})(3^3)}$$

$$\begin{array}{r} 3 \overline{) 243} \\ 3 \overline{) 81} \\ 3 \overline{) 27} \\ 3 \overline{) 9} \\ 3 \overline{) 3} \\ 1 \end{array}$$

$$= \frac{(3^4)^n \cdot \underline{\underline{3^5}} - (3)^{4n-1} \cdot \underline{\underline{3^5}}}{}$$

$$= \frac{(3^2)^{2n} \cdot 3^3 \cdot 3^5 \left( 3^{4n} - 3^{4n-1} \right)}{}$$

$$3^{4n} \cdot 3^3$$

$$\textcircled{1}x + \textcircled{1}x$$

$$\boxed{m \quad n \quad m+n} = \underline{\underline{2x}}$$

Handwritten notes on algebraic rules and properties:

- Rule**  $x \cdot x = x$  (with  $m$  and  $n$  above the  $x$ 's, and  $m+n$  above the result  $x$ , with an arrow pointing from  $m+n$  to the result  $x$ ).
- $x + y \neq xy$  (circled and crossed out with a large red X).
- $x^2 - xy = x(x-y)$  (with an arrow pointing from  $x$  to  $x(x-y)$ ).
- $x^m \cdot x^n = x^{m+n}$  (with  $m$  and  $n$  above the  $x$ 's, and  $m+n$  above the result  $x$ , with an arrow pointing from  $m+n$  to the result  $x$ ).
- $x^m \cdot x^{-n} = x^{m-n}$  (with  $m$  above the first  $x$ ,  $-n$  above the second  $x$ , and  $m-n$  above the result  $x$ , with an arrow pointing from  $m-n$  to the result  $x$ ).

$$= 3^5 \cdot 3^{-3} \left( \underline{\underline{3^{4n}}} - \underline{\underline{3^{4n} \cdot 3^{-1}}} \right)$$

$$= \frac{3^{5-3} \cdot 3^{4n} (1 - 3^{-1})}{}$$

$$= 3^2 \cdot 3^{4n-4n} \left(1 - \frac{1}{3}\right)$$

$$= 928 \cdot 3^0 \left( \frac{1 \times 3}{3} - \frac{1}{3} \right)$$



$$= \cancel{9/5} \times 1 \left( \frac{3-1}{3} \right)$$

$$\textcircled{3} \cancel{9/5} \times \textcircled{2}$$

$$= \textcircled{6}$$

$$= \cancel{\cancel{\cancel{\cancel{\cancel{50}}}}}$$

$$\frac{(243)^{-2/3} (32)^{-1/5}}{\sqrt{(196)^{-1}}}$$

$$= \frac{(3^5)^{-2/3} \times (2^5)^{-1/5}}{\sqrt{\frac{1}{196}}}$$

$$\left[ \begin{array}{l} \left( \frac{a}{b} \right)^{-m} \\ \left( \frac{b}{a} \right)^{tm} \end{array} \right]$$

$$= \frac{3^{-10/3} \times 2^{-1}}{\sqrt{2^2 \times 7^2}}$$

$$\begin{array}{r} 2 \overline{)196} \\ 2 \overline{)98} \\ 7 \overline{)49} \\ 7 \overline{)7} \\ 1 \end{array}$$

$$\sqrt{2^2 \times 7^2}$$

$$\frac{10}{3} \textcircled{3} \sqrt{10}$$

$$= \frac{7^2 \times 1}{3^{-3} \frac{1}{3} \times 2^{-1}} \quad \left| \quad \begin{array}{l} \sqrt[3]{\frac{10}{9}} \\ \frac{1}{1} \\ = \left( 3 + \frac{1}{3} \right) \\ = 3 + \frac{1}{3} \end{array} \right.$$

$$= \frac{1}{3^{3\frac{1}{3}} \times 2^1 \times \frac{1}{2 \times 7}}$$

$$= \frac{1}{3^{3+\frac{1}{3}} \times \cancel{2} \times \frac{1}{\cancel{2} \times 7}}$$

$$= \frac{1}{3^3 \cdot 3^{\frac{1}{3}} \cdot \frac{1}{7}}$$

$$\begin{array}{l} \frac{a}{\frac{b}{c}} \\ a \div \frac{b}{c} \\ \textcircled{a} \times \frac{c}{b} \end{array}$$

$$= \frac{7}{27 \sqrt[3]{3}}$$

$$= \frac{7}{\sqrt[3]{2} \times 27} \quad \left( \begin{array}{l} \boxed{0.17676} \\ 6 \\ \frac{7}{27 \times \sqrt[3]{3}} \end{array} \right)$$

$$L.H.S = \sqrt[3]{3} \times 27 \sqrt{27 \times \sqrt[3]{3}}$$

$$\left( \frac{x^a}{x^b} \right)^{a+b} \times \left( \frac{x^b}{x^c} \right)^{b+c} \times \left( \frac{x^c}{x^a} \right)^{c+a}$$

$$= \left( x^{a-b} \right)^{a+b} \times \left( x^{b-c} \right)^{b+c} \times \left( x^{c-a} \right)^{c+a}$$

$$= \left( x^{a-b} \right)^{a+b} \times \left( x^{b-c} \right)^{b+c} \times \left( x^{c-a} \right)^{c+a}$$

$$= x^{a^2-b^2} \times x^{b^2-c^2} \times x^{c^2-a^2}$$

$$= x^{a^2-b^2+b^2-c^2+c^2-a^2}$$

$$= x^0$$

$$= 1$$

$$= 1 = R.H.S$$

Hence proved  $x^m \cdot y^m = (xy)^m$

Simplify  $\sqrt[3]{\frac{a^l}{a^m}} \times \sqrt[3]{\frac{a^m}{a^n}} \times \sqrt[3]{\frac{a^n}{a^l}}$

$$\begin{aligned}
 &= \sqrt[3]{a^{l-m}} \times \sqrt[3]{a^{m-n}} \times \sqrt[3]{a^{n-l}} \\
 &= \sqrt[3]{a^{l-m} \times a^{m-n} \times a^{n-l}} \\
 &= \sqrt[3]{a^{\cancel{l-m} + m-n + n-l}} \\
 &= \sqrt[3]{a^0} = \sqrt[3]{1} = 1
 \end{aligned}$$

Ans

Simplify

5)  $\left(\frac{a^p}{a^q}\right)^{p+q} \cdot \left(\frac{a^q}{a^r}\right)^{q+r} \div 5(a^p \cdot a^r)^{p-r}, a \neq 0$

6) Simplify  $\left(\frac{a^{2l}}{a^{l+m}}\right) \left(\frac{a^{2m}}{a^{m+n}}\right) \left(\frac{a^{2n}}{a^{n+l}}\right)$

$\dots \frac{2+q}{1} / \frac{q-r}{1} \dots$

$$\begin{aligned}
 S_1 &= (a^{p-q})^{p+q} \cdot (a^{q-r})^{r+p} \div 5(a^{p+r}) \\
 &= a^{p^2-q^2} \cdot a^{q^2-r^2} \div 5a^{p^2-r^2} \\
 &= \frac{a^{p^2-q^2+q^2-r^2}}{5a^{p^2-r^2}} \\
 &= \frac{a^{\cancel{p^2-q^2+q^2-r^2-p^2+r^2}}}{5} \\
 &= \frac{a^0}{5} = \frac{1}{5} A
 \end{aligned}$$

Simplify  $\left(\frac{a^{2l}}{a^{l+m}}\right)\left(\frac{a^{2m}}{a^{m+n}}\right)\left(\frac{a^{2n}}{a^{n+l}}\right)$

$$= a^{2l-l-m} \times a^{2m-m-n} \times a^{2n-n-l}$$

$$\begin{aligned}
 &= a^{l-m} \times a^{m-n} \times a^{n-l} \\
 &= a^{l-m+n-n+l+l-n} \\
 &= a^0 = 1 \quad \checkmark
 \end{aligned}$$

a.

$$\frac{2^{\frac{2}{3}} \cdot 2^{\frac{1}{3}}}{2^{\frac{1}{3}}}$$

b.

$$\begin{aligned}
 &= 2^{\frac{2}{3}} \cdot 2^{\frac{1}{3}-\frac{1}{3}} = 2^{\frac{2}{3}} \cdot 2^0 = 2^{\frac{2}{3}} \quad \checkmark
 \end{aligned}$$

$$\frac{3^{-\frac{1}{2}} \cdot 3^{\frac{1}{2}}}{3^{\frac{1}{2}}}$$

$$= 3^{-\frac{1}{2}} \cdot 3^{\frac{1}{2}} \cdot 3^{-\frac{1}{2}} = 3^{-\frac{1}{2}} \times 3^{\frac{1}{2}-\frac{1}{2}}$$

$$= 3^{-\frac{1}{2}} \times 3^0 = 3^{-\frac{1}{2}} = \frac{1}{3^{\frac{1}{2}}}$$

$$= \frac{1}{\sqrt{3}} \quad \checkmark$$

$$= \sqrt{3}$$

$$\frac{(2x^2)^{-4}}{(-4y^{-1}z^2)^{-2}} = \frac{(-4y^{-1}z^2)^2}{(2x^2)^4}$$

$$(-1 \times 4)^2 =$$

$$= \frac{(-4)^2 \cdot (y^{-1})^2 \cdot (z^2)^2}{2^4 \cdot (x^2)^4} = \frac{2^4 \cdot y^{-2} \cdot z^4}{2^4 \cdot x^8}$$

$$= \frac{2^{4-4} \cdot z^4}{y^2 \cdot x^8} = \frac{z^4}{x^8 y^2}$$

$$\begin{array}{r} 4 \overline{) 28} \\ 2 \overline{) 16} \\ 2 \overline{) 12} \\ 2 \overline{) 8} \\ 2 \overline{) 4} \\ 2 \overline{) 0} \end{array}$$

Simplify the expression

$$\left[ (125)^{\frac{1}{3}} \times (25)^{\frac{1}{2}} + (64)^{\frac{2}{3}} \times 6 + (8)^{\frac{2}{3}} \right]^{\frac{-2}{3}}$$

$$\sqrt[3]{1, 3, \frac{1}{2}, 1, 2, \frac{1}{2}, 1, 3, \frac{2}{3}, 1, 0^3, \frac{2}{3}}^{\frac{-2}{3}}$$

$$= \left[ (5^3)^{\frac{1}{3}} \times (5^2)^{\frac{1}{2}} + (4^3)^{\frac{2}{3}} \times 6 + (2^3)^1 \right]$$

$$= \left[ 5 \times 5 + 4^2 \times 6 + 2^2 \right]^{\frac{-2}{3}}$$

$$= \left[ 25 + 96 + 4 \right]^{\frac{-2}{3}}$$

$$= [125]^{\frac{-2}{3}}$$

$$= 5^{\cancel{3} \times \frac{-2}{\cancel{3}}}$$

$$= 5^{-2} = \frac{1}{5^2} = \frac{1}{25}$$



$$\begin{aligned}
 & \frac{1}{2} \times \frac{2}{1} = 1 \quad \text{① } \frac{1}{16} \times \frac{1}{2} = \frac{1}{32} \\
 & \left[ \left( 2a^{\frac{1}{2}}b^3 \right)^4 \times \left( 4a^{-2}b^{-6} \right)^{-\frac{1}{2}} \right]^{\frac{1}{3}} \\
 & = \left[ 2^4 (a^{\frac{1}{2}})^4 \times (b^3)^4 \times 4^{-\frac{1}{2}} \times (a^{-2})^{-\frac{1}{2}} \times (b^{-6})^{-\frac{1}{2}} \right]^{\frac{1}{3}} \\
 & = \left[ 16 a^2 b^{12} \times (2^{\frac{1}{2}})^{-\frac{1}{2}} \times a^1 \times b^3 \right]^{\frac{1}{3}} \\
 & = \left[ \frac{16}{2} \times a^3 \times b^{15} \right]^{\frac{1}{3}} \\
 & = \left[ 8 \times a^3 \times b^{15} \right]^{\frac{1}{3}} \\
 & = (2^3)^{\frac{1}{3}} \times (a^3)^{\frac{1}{3}} \times (b^{15})^{\frac{1}{3}} \\
 & = 2 \times a \times b^5 \\
 & \boxed{2ab^5}
 \end{aligned}$$

$$= \sqrt{2ab^5}$$

-s-s

1.  $a^r \cdot a^{-s} \div a^s$  is equal to

(a)  $a^{r-s}$

(b)  $a^{r+2s}$

(c)  $a^{r-2s}$

(d)  $a^{-r+2s}$

$$\frac{a^{r-s}}{a^s} =$$

2.  $\sqrt[n]{ab}$  is equivalent to

(a)  $a^{1/n} + b^{1/n}$

(b)  $(ab)^{1/n}$

(c)  $(a+b)^{1/n}$

(d)  $(ab)^n$

3.  $(x^{2/3})^6$  simplifies to

(a)  $x^2$

(b)  $x^3$

(c)  $x^4$

(d)  $x^{12}$

4.  $81^{3/4}$  equals

(a) 9

(b) 27

(c) 81

(d) 243

$$\frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9}$$

$$3^{4 \times 3/4} = 3^3 = 27$$

5.  $16^{-3/4}$  equals

(a) 8

(b)  $\frac{1}{8}$

(c)  $\frac{1}{4}$

(d)  $\frac{1}{16}$

6.  $(\frac{16}{25})^{-1/2}$  equals

(a)  $\frac{4}{5}$

(b)  $\frac{5}{4}$

(c)  $\frac{16}{25}$

(d)  $\frac{25}{16}$

7.  $\sqrt[3]{216}$  equals

(a) 3

(b) 6

(c) 12

(d) 36

8.  $\frac{\sqrt[4]{x^7}}{\sqrt[4]{x^3}}$  simplifies to

(a)  $x$

(b)  $x^2$

(c)  $\sqrt{x}$

(d)  $\sqrt[4]{x}$

$$\frac{\sqrt[4]{x^7}}{\sqrt[4]{x^3}} = \sqrt[4]{\frac{x^7}{x^3}} = (x^4)^{1/4} = x$$

10.  $(p^3q^{-2})^{-1}$  is

(a)  $\frac{p^3}{q^2}$

(b)  $\frac{q^2}{p^3}$

(c)  $p^{-3}q^{-2}$

(d)  $p^3q^2$

11.  $(\sqrt[3]{5})^6$  equals

(a) 5

(b) 15

(c) 25

(d) 125

12.  $32^{2/5}$  equals

(a) 2

(b) 4

(c) 8

(d) 16

$$= p^{-3}q^2 \Rightarrow \frac{q^2}{p^3}$$

$$2^{5 \times \frac{2}{5}} = 2^2 = 4$$

simplify

$$\sqrt[4]{\frac{a^3}{b^3}} \times \sqrt[4]{\frac{b^3}{c^3}} \times \sqrt[4]{\frac{c^3}{a^3}}$$

$$= \sqrt[4]{\frac{\cancel{a^3}}{\cancel{b^3}} \times \frac{\cancel{b^3}}{\cancel{c^3}} \times \frac{\cancel{c^3}}{\cancel{a^3}}}$$

$$= \sqrt[4]{\frac{\cancel{a^2}}{\cancel{b^3}} \times \frac{\cancel{b^3}}{\cancel{c^3}} \times \frac{\cancel{c^3}}{\cancel{a^3}}}$$

$$= \sqrt[4]{1} = 1 \text{ Ans}$$